Autopilots architecture

How a Worker receives work, reasons, acts under its identities, edits documents, and keeps state.

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OPERATING MODEL

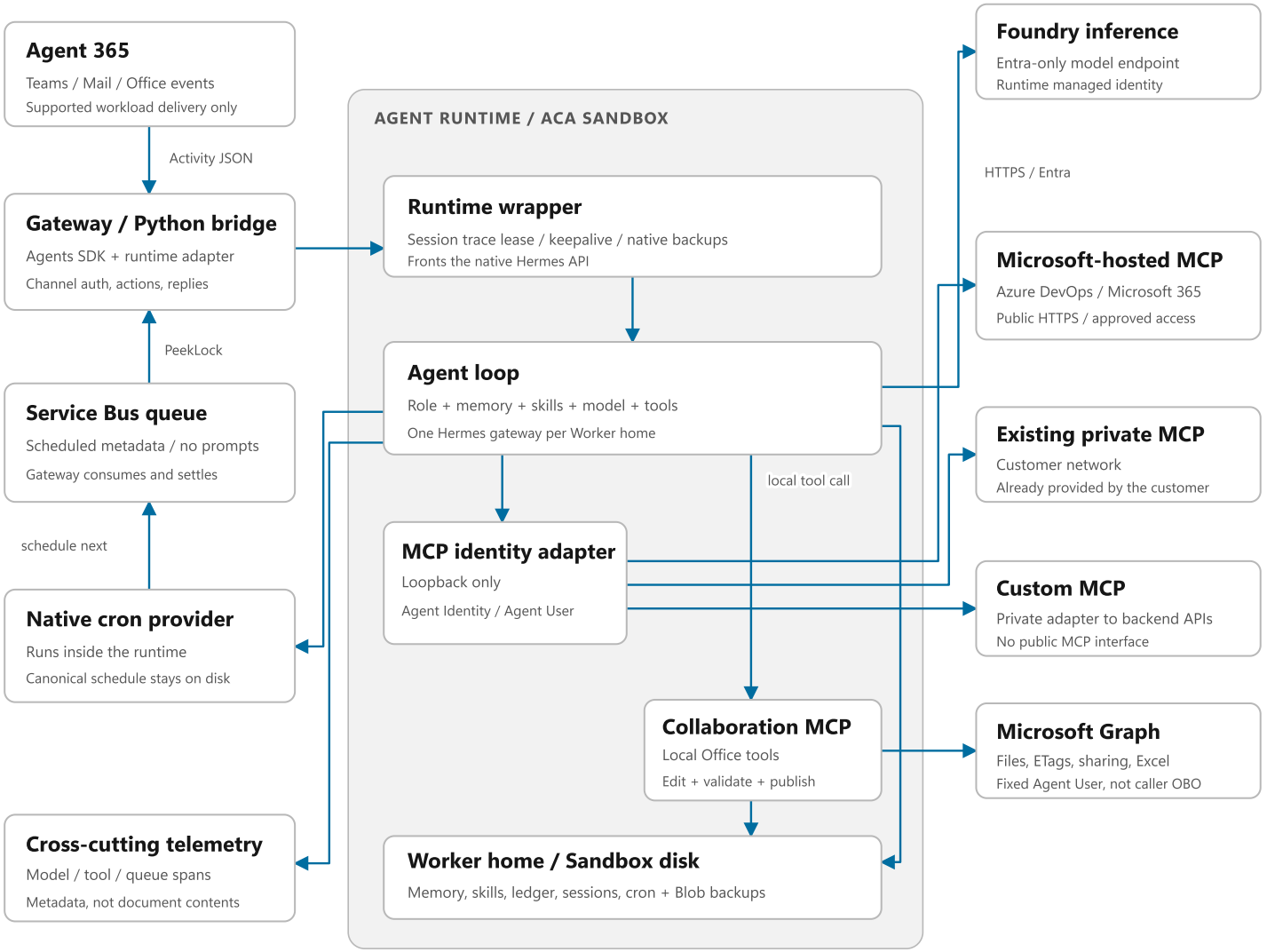
The gateway connects; the Worker does the work.

A **Worker** is a named digital colleague with its own identity, role, private state, and tools. Its **gateway** accepts authenticated activities and calls the runtime URL; deployment owns runtime compute. The **agent runtime** owns reasoning, tool use, memory, skills, and document work. Foundry supplies model inference; it does not host the agent's compute.

The runtime runs Hermes 0.22.5 from the official image, pinned by digest. A runtime wrapper adds trace correlation, keepalive, and native backups around the agent loop; the `autopilots` and `azure` Hermes plugins add keepalive, telemetry, attachment privacy, internal routes, and the Service Bus cron provider. Model and tool traffic leaves the runtime directly; the gateway is neither an inference proxy nor a general tool router.

Solid: connection in use Dashed: optional connection or component

01 / Application responsibilities and interfaces



The cron box is drawn outside for clarity; its code executes in the runtime.

Arrows show calls or trigger delivery. Tool access follows the three integration patterns in the infrastructure view; telemetry spans the stack.

01

Five service roles, separate Sandbox Groups.

Each role has its own Sandbox Group and managed identity.

GATEWAY

Receives Agent 365 activities and scheduled work.

RUNTIME

Runs Hermes against its Worker home, backed up to a Blob volume.

PRIVATE MCP

Serves incidents through private networking.

PUBLIC MCP

Serves authenticated shipments over public HTTPS.

GENERATED APPS

Runs Worker-created applications outside the agent runtime.

Each Group has its own user-assigned managed identity. Foundry supplies inference and external-agent monitoring, not compute. An Express managed environment supplies the private MCP networking foundation.

> SOURCE
>
> Platform infrastructure · Worker infrastructure · Service deployment

CONVERSATION TURNS

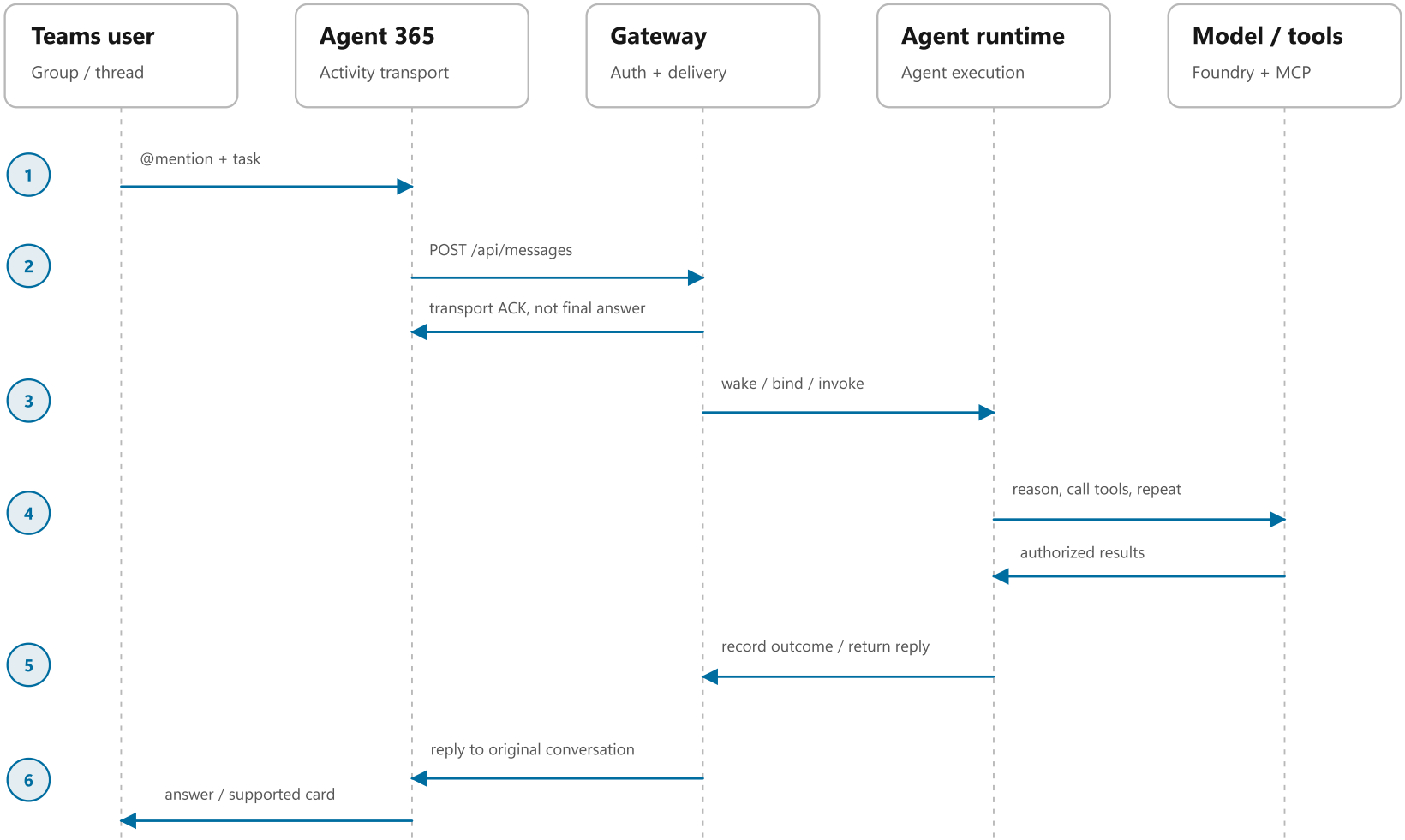
02

Acknowledge transport; finish the turn in the runtime.

The HTTP acknowledgement is not the business result.

A user explicitly mentions the Agent User in a group conversation, and the host delivers the activity to the registered Worker endpoint. The gateway does not subscribe to all channel traffic and cannot assume complete group history.

02 / Interactive Teams request



Service Bus enters only when the task creates scheduled work or a publication retry.

The order illustrates one supported interaction, not an exact packet capture or latency promise. The model/tool loop can take several calls. Public group output must respect the conversation's audience.

1. **Authenticate before invoking the Worker**
   The Agents SDK validates Activity Protocol authentication. The bridge resolves the Worker, caller, and conversation; unsupported or invalid input does not become an agent task.

2. **Reuse state without allowing concurrent writers**
   Deployment creates the runtime; the gateway request wakes it through OnDemand ingress. The wrapper binds trace context to the native session and keeps compute awake while the request runs; Hermes reads the Worker home and performs the requested work.

3. **Reply through the originating workload**
   The bridge delivers the result through the channel. Office comment requests stay in their comment thread. An early HTTP acknowledgement is not a durable queue entry or proof of task completion.

Email and Office notifications reply in their originating workload rather than being redirected to Teams.

> SOURCE
>
> Gateway and runtime adapter · Wrapper and native runtime integration

IDENTITY AND NETWORK

03

Choose the principal for the resource.

Managed identity, Agent Identity, and Agent User have distinct jobs.

A managed identity runs Azure work, the Agent Identity calls autonomous tools, and the Agent User holds the Worker's Microsoft 365 account. A network route never substitutes for one of these decisions.

CALL	PRINCIPAL	ENFORCEMENT
Gateway to runtime	Operator API key	Runtime wrapper over OnDemand ingress; the gateway has no runtime Sandbox Group role.
Gateway to generated-app lifecycle	Gateway managed identity	Sandbox Group data-plane role on the generated-apps Group.
Agent runtime to Foundry	Runtime managed identity	Entra token and model-resource RBAC through the native provider.
Agent runtime to custom MCP	Agent Identity	Expected tenant, issuer, audience, caller, and application role.
Work IQ and Microsoft Graph	Fixed Agent User via Agent Identity exchange	Approved delegated permissions and resource sharing. Not human on-behalf-of access.
Runtime cron to Service Bus	Agent Identity	Queue-scoped Data Sender role.
Gateway queue receive and document retry	Gateway managed identity	Queue-scoped Data Receiver and Data Sender roles.

<details>
<summary>Inspect the token path</summary>

The loopback MCP adapter obtains a workload assertion, exchanges through the blueprint federated credential using `fmi_path`, and requests the target audience. Microsoft 365 access adds the fixed Agent User exchange through `user_fic`. Tokens are short-lived; the agent does not receive bearer-token text. This is not human on-behalf-of delegation.
</details>

SOURCE

Token exchange and MCP adapter · Worker role assignments

04

Private networking and Entra checks both apply.

A public MCP endpoint is not anonymous; a private one is not automatically trusted.

Hermes connects to loopback MCP endpoints. The adapter selects the business identity and upstream route. Private incidents use the Sandbox network connection, peering, private DNS, and a linked private endpoint; public shipments use HTTPS.

Both custom servers validate tokens for their configured tenant, issuer, audience, caller, and required role. Network reachability does not grant application access. The gateway is not a relay for this tool traffic.

SOURCE

Network topology · Private server · Public server

DOCUMENTS

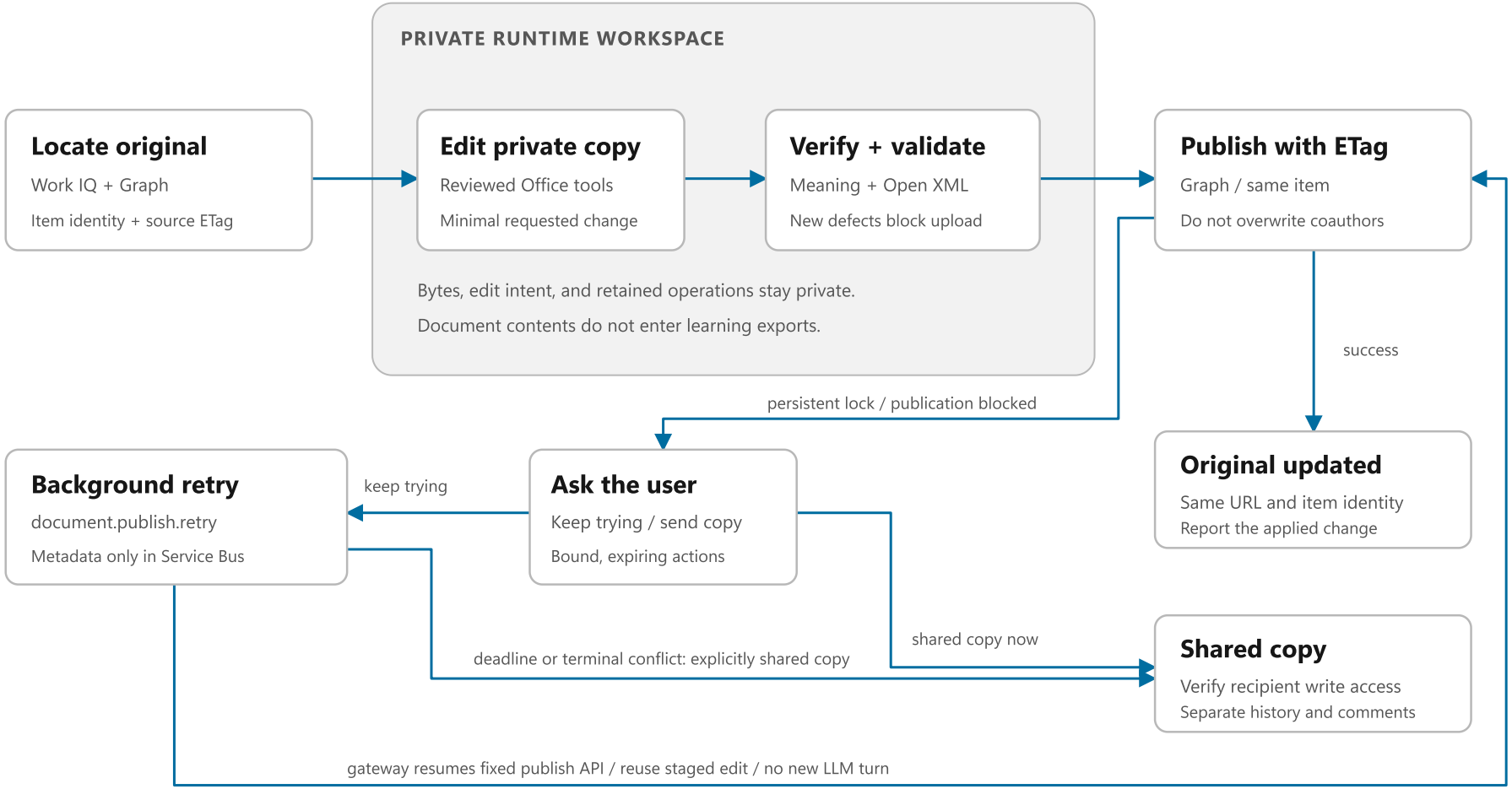
05

Edit a private copy; publish to the same item.

ETag conflicts and locks are explicit outcomes, not silent overwrites.

Work IQ handles supported discovery, reading, creation, and comments. For a Word body edit, reviewed Office tools inside the runtime change a private copy, and the collaboration MCP coordinates Graph download, validation, and same-item publication.

03 / Document edit and publication lifecycle



A changed source is re-read. Only reconstructible guarded edits can rebase; opaque edits fail closed.

Preparation and model reasoning happen once. A background publication retry executes a storage workflow, not a repeated “try again” prompt.

Same-item publication preserves the document's identity, sharing, comments, and history. A copy is reported as a different item. The Worker never reports the original as updated when the upload failed.

> Inspect locks and retry behavior

A locked edit is staged privately for one hour by default. Choosing background publication starts a separate 24-hour retry deadline and retains the file for one further hour for completion/cleanup. Teams actions select retrying the original or sharing a copy; retries do not start another model turn. See the document contract for configuration and receipt retention.

ETag conflicts are not silently overwritten. Attachment content is turn-scoped and excluded from durable learning; pending publication is a separate bounded retention path. External message delivery is not exactly-once.

SOURCE

Collaboration and file boundary · Document and attachment limits

STATE AND RECOVERY

06

The Worker home carries continuity; native backups make it durable.

Images provide software; the Hermes home provides identity-specific state.

The home, `/data/hermes` on the runtime Sandbox root disk, contains native memory and history, private skills, governed role/candidate skills, the skill ledger and owner decisions, schedules, and working files. Hermes 0.22.5 runs one gateway per home, so the Worker is the home itself rather than a named profile. A pinned Role Release provides the baseline against which local governed changes are tracked.

After five idle minutes, compute suspends to memory, so an OnDemand request resumes the same Sandbox ID with Hermes still running.

The wrapper checks the home once a minute and, after a change, runs native `hermes backup`; logs, caches, heartbeats, and locks that idle Hermes rewrites do not count. The archive is staged locally, then copied and renamed onto the Worker's `hermes-backups` AzureBlob Sandbox volume at `/backup`, which keeps the newest 24. A runtime that starts with an empty home restores the newest archive with native `hermes import`, checks the Worker ID, and stops on a damaged or foreign archive.

EVENT	DATA KEPT
Idle suspend, resume, platform maintenance	Everything, including running processes
Image, setting, or Role Release change	Everything up to the final backup deployment runs before deleting the old runtime
`Failed` or lost Sandbox	Everything up to the newest automatic backup; at most about the last minute of changes is lost
Deleted Sandbox Group	Nothing; the backup volume belongs to the group

A Role Release refresh is a separate governed operation, not an automatic side effect of a new image. `/health` reports the last published backup and any backup error.

> Inspect why the home is not on a Data Disk

A Data Disk volume allows only Disk suspend, which restarts Hermes on every wake, including eager discovery of eleven MCP servers. Measured on 2026-10-31 in Sweden Central with 2 vCPU and 2 GiB, Disk mode needed 17–25 s to the native Hermes API; a Memory resume needed 1.5–2.4 s.

A live home on the Blob volume fails because SQLite snapshots and locking fail on blobfuse. Sandbox snapshots restore the old image. A customer-owned Storage account would need a Private Endpoint, private DNS, and RBAC because subscription policy forces private networking.

SOURCE

Backup and restore · Runtime deployment and final backup · Memory suspend decision · Refresh rules

07

Service Bus carries wake-up identifiers, not prompts.

The runtime owns schedule meaning; the gateway owns queue reception.

Persist schedule
The Worker home retains the job, revision, and next occurrence.

Receive due work
The awake gateway takes a PeekLock and wakes the runtime.

Execute and settle
Validate revision, record outcome, deliver, and complete.

MESSAGE TYPE	PRODUCER / CREDENTIAL	CONSUMER ACTION
`hermes.cron.fire`	Runtime cron provider / federated Agent Identity	Claim the due occurrence, run the saved task, record output, deliver to its bound conversation, and reconcile the next occurrence.
`system.dream`	Runtime cron provider / federated Agent Identity	Run the reserved reflection workflow with fenced checkpoints. Ambiguous already-started work stops for inspection rather than blindly running again.
`document.publish.retry`	Gateway / gateway managed identity	Resume a saved publication operation through a fixed API. No new model turn is needed.

User tasks, the system Dream, and document retries share this transport. Durable claims and receipts distinguish retryable work from an ambiguous prior execution; failures stay visible.

SOURCE

Cron store integration · Dream phases · Queue consumer and delivery

08

Recover known state; stop on unknown outcomes.

Lifecycle recovery must not become a blind replay of business actions.

- **Stopped service:** resume the matching Sandbox instead of replacing it.
- **Failed gateway or MCP:** create a healthy replacement before retiring the prior resource.
- **Failed runtime:** rerun deployment; the replacement restores the newest automatic backup.
- **Unknown Dream outcome:** surface the fenced phase; do not rerun the model automatically.
- **Changed governed state:** require a new packet and approval, not an old receipt.
- **Missing endpoint ownership:** fail preflight before Agent 365 registration is mutated.

SOURCE

Service recovery · Runtime replacement · Dream recovery · Endpoint registration

Keep the gateway thin, the Worker home backed up, and every authorization and recovery decision explicit.

▼ Agent 365 channel and identity binding

Channel identity

Each Worker has an Agent Identity and Agent User. Agent 365 binds its endpoint and permissions, then delivers Microsoft 365 activities to `/api/messages`. The Agents SDK authenticates incoming requests.

▼ Gateway Sandbox

Always awake

The gateway authenticates activities, binds conversations, calls the runtime URL (OnDemand ingress wakes compute), consumes Service Bus, and delivers replies. It stays awake for queued and post-acknowledgement work. Model calls and MCP traffic go directly from the runtime.

▼ Runtime wrapper

Runtime-local boundary

The wrapper fronts the native Hermes API. It binds trace context to explicit native sessions, keeps compute awake during mutating requests, serves health and document routes, and publishes native backups. An interrupted trace lease returns `409` instead of misattributing later work. Learning itself runs natively in Hermes.

▼ Agent loop

Agent behaviour

The agent runtime combines role, memory, and skills; calls the model; and selects authorized tools until it can respond. Private learned state stays with the Worker. Reasoning does not run in the gateway.

▼ Loopback MCP identity adapter

Runtime-local authentication

The adapter refreshes tokens for approved MCP endpoints. Autonomous tools use Agent Identity; supported Microsoft 365 calls use the fixed Agent User. It is not human on-behalf-of access or a model proxy.

▼ Collaboration MCP and reviewed Office tools

Private local editing

A loopback MCP wraps reviewed Office tools and Graph publication. It edits a private copy, checks meaning, and validates Open XML. New defects block upload; tooling is built into the runtime image.

▼ Worker home

Sandbox disk with native backups

The Hermes home `/data/hermes` on the Sandbox root disk holds memory, skills, the skill ledger, sessions, cron, receipts, and document staging. Memory suspend and resume preserve it; native backups on the `hermes-backups` Blob volume carry it into a replacement runtime. The home belongs to the Worker, not each caller. Raw documents and private state stay out of shared learning exports.

▼ Service Bus queue and dead-letter handling

Per-Worker queue

The queue carries cron, Dream, and document-retry triggers, never prompts or document contents. The awake gateway receives with PeekLock and settles each message explicitly. Runtime cron sends as Agent Identity; gateway receive and retry send use the gateway managed identity.

▼ Native cron provider

Scheduled work

Schedules stay in the Worker home and travel with its backups. Cron sends the next occurrence to Service Bus; the gateway wakes the agent runtime when due. Proactive delivery requires a previously bound conversation. Scheduling is enabled per Worker.

▼ Observability and correlation

Cross-cutting metadata

Gateway, queue, model, and tool spans report timing, correlation, and usage to Azure Monitor. Default traces exclude prompts and document contents. Foundry registers the external agent; some platform hops retain correlation gaps.

▼ Foundry model inference

Model API

The agent runtime calls Foundry directly through the native provider. Its managed identity and RBAC authorize inference; local authentication is disabled. The gateway is not an inference proxy.

The reference deployment uses the Entra-only public endpoint; the production target adds a private endpoint.

▼ Microsoft-hosted MCP tools

Public HTTPS

Use approved Microsoft-hosted tools, such as Azure DevOps and Microsoft 365 MCP integrations. Configure permissions for each service. Microsoft 365 operations use the approved Agent User flow; other services require their own supported connection.

▼ Existing private MCP

Customer-provided service

Connect to an MCP server already running in the customer network through private routing and DNS. Reuse its approved authentication and permissions; no new MCP host is needed in this solution.

▼ Custom MCP

Optional adapter

Add an ACA-hosted adapter when a backend lacks MCP. It exposes a private MCP interface and reaches backend APIs through the workload VNet and enterprise hub. Use a supported private-ingress hosting configuration.

▼ Microsoft Graph under the Agent User

Deterministic file operations

Graph handles downloads, ETags, guarded uploads, sharing, and supported Excel writes. Calls use the fixed Agent User's approved permissions. Verify the recipient's access before delivering an output link.

▼ Teams and Microsoft 365 entry points

Reference implementation

Personal messages and explicit Agent User mentions are supported entry points. Delivered replies, selected reactions, Mail notifications, and Office comment events have workload-specific handling.

Shared groups are a shared audience. Private per-person delivery and human on-behalf-of identity are not implemented.

▼ Agent runtime

ACA Sandbox

Deployment creates agent compute; gateway requests wake it through OnDemand ingress. Hermes 0.22.5 keeps the Worker home on the Sandbox disk, backed up natively to a Blob volume; one Hermes gateway writes it. The `autopilots` plugin's keepalive protects active execution. Five idle minutes suspend compute to memory, and resume continues the running processes with the same Sandbox ID.

▼ Model and tool loop

Illustrated runtime behaviour

The agent runtime calls Foundry for reasoning and invokes authorized MCP/native tools as needed. These are separate calls with separate credentials and failure modes. An external tool can be private-network reachable yet still deny access.

The example is a functional sequence, not measured call counts, timing, or an assurance that every request needs every tool.

▼ Locate and download the original

Reference document workflow

Work IQ helps discover the shared item and its meaning. Deterministic Graph access captures the source item identity and ETag and downloads the document into a private Worker workspace. A link alone is not proof of permission.

▼ Edit a private copy

Reference document workflow

Reviewed local Office tools apply only the requested change. Source bytes, excerpts, staged output, edit intent, and operation metadata remain private. These are not stored in Service Bus and are not exported as shared skills.

▼ Verify meaning and package validity

Reference document workflow

Verify the requested semantic change and validate Open XML against the Microsoft 365 target. Existing validation defects are a private baseline; newly introduced defects prevent publication. Package validity alone does not prove that the edit means what the user asked.

▼ ETag-guarded same-item publication

Reference document workflow

Publish to the original item using its source ETag. If the source changed, re-read it. Only reconstructible guarded edits may rebase; opaque transformations fail closed instead of overwriting coauthor changes.

Apply the body update before claiming success in comments or messages. A lock does not justify checking out the document to block other authors.

▼ Deterministic background publication

Reference document workflow / no repeated model call

The gateway schedules `document.publish.retry` with an opaque operation ID and later invokes a fixed runtime publication API. Bytes, edit intent, source ETag, recipient, and delivery reference stay in the Worker home.

The checked-in retry choice uses a 24-hour deadline. At expiry or a terminal rebase conflict it produces an explicitly shared copy, not a false “original updated” result. Receipts protect state transitions but cannot make external delivery exactly-once. The duration is an implementation policy, not an Azure limit.

▼ Explicit user choice after a persistent lock

Reference document workflow

Offer reviewed actions: “Keep trying original” or “Send shared copy now”. Authenticated encrypted tokens bind the Worker, user, conversation, operation, and expiry; the gateway consumes actions idempotently. The agent does not invent raw action payloads.

▼ Original item updated

Reference document workflow

A successful same-item write preserves item identity, URL, sharing context, comments, and version history. Record the mutation receipt and report the applied change. When the request originated in an Office comment, reply in that thread rather than an unrelated Teams chat.

▼ Explicitly shared copy

Reference document workflow / distinct outcome

Create a separate Agent User-owned copy and verify the intended user's write permission before delivering its link. An unshareable copy is removed. The response must state that the original was not changed and that the copy has separate comments, sharing, and history.